Ex No.:3
Date:

Poorna Sri.A
953625148038

1. Aim:

To write a C++ program to read and display the elements of a one-dimensional array.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of elements in the array.
Step 3: Declare an array of the given size.
Step 4: Read the array elements.
Step 5: Display all the array elements.
Step 6: Stop the program.

Program Code:

```cpp
#include<iostream>
using namespace std;
int main(){
   int n;
   cout<<"Enter the no.of elements:";
   cin>>n;
   int arr[n];
   cout<<"Enter the array elements:";
   for(int i=0;i<n;i++){
      cin>>arr[i];
   }
   cout<<"Array Elements:";
   for(int i=0;i<n;i++){
      cout<<arr[i]<<" ";
   }
   return 0;
}
```

Sample Output:

```
Enter the no.of elements:5
Enter the array elements:1 2 3 4 5
Array Elements:1 2 3 4 5

=== Code Execution Successful ===
```

Result:

Thus, the C++ program to read and display the elements of an array was executed successfully.

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2. Aim:

To write a C++ program to search for an element in an array using linear search.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of elements.
Step 3: Read the array elements.
Step 4: Read the element to be searched.
Step 5: Compare the search element with each array element.
Step 6: If the element is found, display its position.
Step 7: Otherwise, display "Element Not Found".
Step 8: Stop the program.

Program Code:

```
#include<iostream>
using namespace std;
int main(){
    int n,key,found=0;
    cout<<"Enter the no.of elements:";
    cin>>n;
    int arr[n];
    cout<<"Enter the array elements:";
    for(int i=0;i<n;i++){
        cin>>arr[i];
    }
    cout<<"Enter the array element to be searched:";
    cin>>key;
    for(int i=0;i<n;i++){
        if(arr[i]=key){
            cout<<"Element found a position:";
            found=1;
            break;
        }
    }
    if(found=0){
        cout<<"Element Not Found";
    }
    return 0;
}
```

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Sample Output:

```
Enter the no.of elements:5
Enter the array elements:1 2 3 4 5
Enter the array element to be searched:3
Element found  in a position:

=== Code Execution Successful ===
```

Result:

Thus, the C++ program to search for an element in an array using linear search was executed successfully.

3. Aim:

To write a C++ program to display the elements of an array in reverse order.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of elements.
Step 3: Read the array elements.
Step 4: Traverse the array from the last element to the first element.
Step 5: Display the elements in reverse order.
Step 6: Stop the program.

Program Code:

```
#include<iostream>
using namespace std;
int main(){
   int n;
   cout<<"Enter the no.of elements:";
   cin>>n;
   int arr[n];
   cout<<"Enter the array elements:";
   for(int i=0;i<n;i++){
      cin>>arr[i];
   }
   cout<<"Array in the reverse order:";
   for(int i=n-1;i>=0;i--){
      cout<<arr[i];
   }
   return 0;
}
```

Sample Output:

```
Enter the no.of elements:5
Enter the array elements:1 2 3 4 5
Array in the reverse order:54321

=== Code Execution Successful ===
```

Result:

Thus, the C++ program to display the array elements in reverse order was executed successfully.

4. Aim:

To write a C++ program to find the smallest and largest elements in an array.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of elements.
Step 3: Read the array elements.
Step 4: Assume the first element as the smallest and largest.
Step 5: Compare each remaining element with the smallest and largest values.
Step 6: Update the smallest and largest values whenever required.
Step 7: Display the smallest and largest elements.
Step 8: Stop the program.

Program Code:

```
#include<iostream>
using namespace std;
int main(){
   int n;
   cout<<"Enter the no.of elements:";
   cin>>n;
   int arr[n];
   cout<<"Enter the array elements:";
   for(int i=0;i<n;i++){
      cin>>arr[i];
   }
   int smallest=arr[0];
   int largest=arr[0];
   for(int i=1;i<n;i++){
      if(arr[i]<smallest){
         smallest=arr[i];
```

```
        }
        if(arr[i]>largest){
            largest=arr[i];
        }
    }
    cout<<"Smallest Element:"<<smallest<<endl;
    cout<<"Largest Element:"<<largest<<endl;
    return 0;
}
```

Sample Output:

```
Enter the no.of elements:5
Enter the array elements:1  9 5 7 3
Smallest Element:1
Largest Element:9


=== Code Execution Successful ===
```

Result:

Thus, the C++ program to find the smallest and largest elements in an array was executed successfully.

5. Aim:

To write a C++ program to merge two arrays into a single array.

Algorithm:

Step 1: Start the program.
Step 2: Read the size of the first array.
Step 3: Read the size of the second array.
Step 4: Read the elements of both arrays.
Step 5: Copy the elements of the first array into the merged array.
Step 6: Copy the elements of the second array after the first array elements.
Step 7: Display the merged array.
Step 8: Stop the program.

Program Code:

```
#include<iostream>
using namespace std;
int main(){
    int n1,n2;
    cout<<"Enter the size of the first array:";
    cin>>n1;
```

```
int arr1[n1];
cout<<"Enter the size of the second array:";
cin>>n2;
int arr2[n2];
cout<<"Enter the first array elements:";
for(int i=0;i<n1;i++){
   cin>>arr1[i];
}
cout<<"Enter the second array elements:";
for(int i=0;i<n2;i++){
   cin>>arr2[i];
}
int merge[n1+n2];
for(int i = 0; i < n1; i++) {
   merge[i] = arr1[i];
}
for(int i = 0; i < n2; i++) {
merge[n1 + i] = arr2[i];
}
cout << "Merged array: ";
for(int i = 0; i < n1 + n2; i++) {
   cout << merge[i] << " ";
}
return 0;
}
```

Sample Output:

```
Enter the size of the first array:3
Enter the size of the second array:3
Enter the first array elements1 2 3
Enter the second array elements4 5 6
Merged array: 1 2 3 4 5 6

=== Code Execution Successful ===
```

Result:

Thus, the C++ program to merge two arrays into a single array was executed successfully.

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6. Aim:

To write a C++ program to find the sum of all elements in an array.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of elements.
Step 3: Read the array elements.
Step 4: Initialize the sum variable to zero.
Step 5: Add each array element to the sum.
Step 6: Display the sum of the array elements.
Step 7: Stop the program.

Program Code:

```
#include<iostream>
int main(){
    int n,i,sum=0;
    printf("Enter size of arra:");
    scanf("%d",&n);
    int a[n];
    printf("Enter the array elements:");
    for(int i=0;i<v;i++){
        scanf("%d",&a[i]);
        sum+=a[i];
    }
    printf("Sum:%d",sum);
    return 0;
}
```

Sample Output:

```
Enter the no.of elements:5
Enter the array elements:1 2 3 4 5
Sum=15

=== Code Execution Successful ===
```

Result:

Thus, the C++ program to find the sum of all elements in an array was executed successfully.

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7. Aim:

To write a C++ program to read and display the elements of a two-dimensional array (matrix).

Algorithm:

Algorithm:
Step 1: Start the program.
Step 2: Read the number of rows and columns.
Step 3: Declare a two-dimensional array.
Step 4: Read the matrix elements.
Step 5: Display the matrix elements row by row.
Step 6: Stop the program.

Program Code:

```
#include <iostream>
using namespace std;
int main() {
  int r, c;
  cout << "Enter rows and columns: ";
  cin >> r >> c;
  int arr[r][c];
  cout << "Enter the elements:";
  for (int i = 0; i < r; i++) {
    for (int j = 0; j < c; j++) {
      cin >> arr[i][j];
    }
  }
  cout << "2D Array:\n";
  for (int i = 0; i < r; i++) {
    for (int j = 0; j < c; j++) {
      cout << arr[i][j] << " ";
    }
    cout << endl;
  }
  return 0;
}
```

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953625148038

Sample Output:

```
Enter rows and columns: 3 3
Enter the elements:1 2 3 4 5 6 7 8 9
2D Array:
1 2 3
4 5 6
7 8 9


=== Code Execution Successful ===
```

Result:

Thus, the C++ program to read and display the elements of a two-dimensional array was executed successfully.

8. Aim:

To write a C++ program to find the transpose of a matrix.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of rows and columns.
Step 3: Read the matrix elements.
Step 4: Interchange the rows and columns while displaying the matrix.
Step 5: Display the transpose of the matrix.
Step 6: Stop the program.

Program Code:

```
#include <iostream>
using namespace std;
int main() {
   int r, c;
   cout << "Enter rows and columns: ";
   cin >> r >> c;
   int arr[r][c];
   cout << "Enter the elements:";
   for (int i = 0; i < r; i++) {
      for (int j = 0; j < c; j++) {
         cin >> arr[i][j];
      }
   }
   cout << "Transpose Matrix:\n";
   for (int i = 0; i < c; i++) {
```

```
        for (int j = 0; j < r; j++) {
            cout << arr[j][i] << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Sample Output:

```
Enter rows and columns: 3 3
Enter the elements:1 2 3 4 5 6 7 8 9
Transpose Matrix:
1 4 7
2 5 8
3 6 9


=== Code Execution Successful ===
```

Result:

Thus, the C++ program to find the transpose of a matrix was executed successfully.

9. Aim:

To write a C++ program to display the elements of a matrix in reverse order.

Algorithm:

Step 1: Start the program.
Step 2: Read the number of rows and columns.
Step 3: Read the matrix elements.
Step 4: Traverse the matrix from the last row and last column towards the first row and first column.
Step 5: Display the matrix elements in reverse order.
Step 6: Stop the program.

Program Code:

```
#include <iostream>
using namespace std;
int main() {
    int r, c;
    cout << "Enter rows and columns: ";
    cin >> r >> c;
    int arr[r][c];
    cout << "Enter the elements:";
    for (int i = 0; i < r; i++) {
```

```
        for (int j = 0; j < c; j++) {
            cin >> arr[i][j];
        }
    }
    cout <<"Reversed Matrix:\n";
    for (int i = r-1; i>=0; i--) {
        for (int j =c-1; j >=0; j--) {
            cout << arr[j][i] << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Sample Output:

```
Enter rows and columns: 3 3
Enter the elements:1 2 3 4 5 6 7 8 9
Reversed Matrix:
9 6 3
8 5 2
7 4 1


=== Code Execution Successful ===
```

Result:

Thus, the C++ program to display the elements of a matrix in reverse order was executed successfully.